



Munich University of Applied Sciences

Hochschule Muenchen

Department of Computer Science & Mathematics

Fakultaet fuer Informatik und Mathematik

ARIA-NBV: Target-Conditioned, Quality-Driven Next-Best-View Planning

ARIA-NBV: Zielkonditionierte, qualitaetsgetriebene Next-Best-View-Planung

Master's Thesis Proposal / Proposal zur Masterarbeit

for obtaining the academic degree Master of Science (M.Sc.)

submitted to
Munich University of Applied Sciences
Department of Computer Science & Mathematics

Submitted by: Jan Duchscherer
Email: j.duchscherer@hm.edu
Program: Informatik - Master
Specialization: Visual Computing and Machine Learning
Matriculation Number: 00807819

First Examiner: Prof. Dr. Markus Friedrich
Supervisor: Prof. Dr. Markus Friedrich
Submission Date: 30 April 2026

Transparency in the Use of AI Tools

AI-assisted tools were used to organize literature notes, check consistency across repository documentation, and draft parts of the proposal text. The author remains responsible for the final research scope, technical claims, citations, and submitted document.

Contents

1	Motivation	1
2	Problem and Research Contract	2
3	Related Work and Positioning	4
4	Objectives and Hypotheses	8
5	Method	9
5.1	M1/M2: Data, Geometry, and One-Step Baseline	9
5.2	M3/M4: Target Contract and Target-Conditioned Scoring	10
5.3	M5: Bounded Rollout and Q_H Training	10
5.4	Evaluation	11
6	Schedule and Risk Control	12
7	Preliminary Thesis Outline	14
	Bibliography	15

1 Motivation

Active reconstruction is a coupled sensing and inference problem: the reconstruction error after a fixed budget depends as much on the selected viewpoints as on the downstream surface estimator. Classical active perception therefore treats sensing actions as part of perception itself, and view-planning surveys formalize the dominant generate-score-select loop for three-dimensional inspection [AWB88, Baj88, SRR03]. ARIA-NBV adopts that loop for egocentric indoor data, but asks a narrower scientific question: can finite candidate views be scored and planned by reconstruction-quality improvement rather than by coverage or uncertainty proxies alone?

The key empirical precedent is VIN-NBV, which replaces pure coverage with Relative Reconstruction Improvement (RRI), an oracle label computed from the reduction in Chamfer-style reconstruction error after adding a query view [Fra+25]. This is the right axis for ARIA-NBV because target indoor surfaces can remain poorly reconstructed even when many voxels or pixels are already covered. The thesis extends this idea from object-centric RGB-D benchmarks to the Project Aria / ASE ecosystem, where calibrated egocentric streams, trajectories, semi-dense points, predicted object boxes, and a mesh-supervised subset support controlled oracle labels [Eng+23, Met25, Str+24a].

The current gap is not a lack of possible algorithms; it is the lack of a trustworthy target-conditioned, multi-step evidence chain. GenNBV and Hestia show that continuous 5-DoF and hierarchical NBV policies are plausible when an online simulator and coverage reward are mature [Che+24, Lu+26]. Radiance-field and Gaussian-splatting papers show that view utility can be decomposed into uncertainty, Fisher information, semantics, dynamics, object identity, or downstream task error [BLL25, Jeo+26, JLD24, LJD25, Pan+22, Str+24b]. ARIA-NBV should use those papers to sharpen the model, not to replace its objective: the thesis utility remains scene and target RRI, with validity and cost reported as separate constraints.

THESIS POSITION

The thesis contribution is a target-conditioned, quality-driven finite-candidate NBV study on ASE/EFM. The hard result is a candidate-query Q_H model that predicts bounded cumulative target RRI over valid candidates and improves on one-step greedy/model scoring under the same acquisition budget. Continuous actor-critic control, Gymnasium/SB3, Habitat/Isaac, SceneScript-style global memory, and real-device guidance are bridge or future work unless this finite-candidate result is already stable.

2 Problem and Research Contract

ARIA-NBV is formulated as a finite-candidate, target-conditioned NBV problem. At rollout step t , the actor-visible counterfactual state is

$$s_t^{\text{cf0}} = (F_v^{\text{root}}, P_t, Q_t, m_t, \rho_t, z_e, b_t)$$

where F_v^{root} is the frozen EVL/EFM voxel-field context at the snippet root, P_t is accumulated semi-dense or rendered/fused geometry, $Q_t = \{q_{t,i}\}_{i=1}^{N_q}$ is the finite candidate table, $m_{t,i} \in \{0, 1\}$ is the validity mask, $\rho_{t,i}$ is the invalidity reason, z_e is the actor-visible descriptor of the selected target e , and b_t is the remaining acquisition budget. The oracle-only state augments this with the ASE GT mesh M and target mesh crop M_e ; these are label/evaluation assets and are not actor inputs in the main V1 protocol.

Target-conditioned evaluation uses OBS-SEL / PRED-Q / GT-EVAL. Target selection uses only observed or predicted evidence. Scoring and Q_H receive the predicted/observed target descriptor z_e . Labels and final evaluation are computed by matching that actor-visible target to a GT target crop M_e . V0 may use GT boxes as a sanity or upper-bound path, but V1 is the main thesis claim.

For a target crop operator $C_{e(\cdot)}$, define the target distance

$$D_{e(P)} = \text{CD}(C_{e(P)}, M_e),$$

with the same accuracy/completeness surface-distance family used by the shared RRI equations. If action $a_t = i$ selects valid candidate $q_{t,i}$, the counterfactual geometry transition is

$$P_{t+1} = P_t \cup P_{q_{t,i}}.$$

The immediate target reward is state-relative target RRI:

$$r_t^e = \frac{D_{e(P_t)} - D_{e(P_{t+1})}}{D_{e(P_t)} + \varepsilon}.$$

The endpoint quality metric for a rollout sequence $\tau = (a_0, \dots, a_{H-1})$ is instead root-relative:

$$J_e^H(\tau) = \frac{D_{e(P_0)} - D_{e(P_H)}}{D_{e(P_0)} + \varepsilon}.$$

The learning return is additive:

$$G_t^H = \sum_{k=0}^{H-1} \gamma^k r_{t+k}^e.$$

Thus G_t^H trains the finite-horizon value function, while J_e^H reports the fraction of initial target error removed at the fixed budget. Negative target RRI remains a valid signal when a view worsens target distance; invalid actions are constraints and are hard-masked before argmax, softmax, loss, and bootstrap operations.

The central research question is:

MAIN QUESTION

Can ARIA-NBV train a target-conditioned candidate-query Transformer $Q_{H,\theta}(s_t^{\text{cf0}}, z_e, q_{t,i})$ that predicts one masked bounded-horizon value per finite candidate and whose selected actions, when re-evaluated by the ASE oracle, beat one-step greedy/model scoring on cumulative target RRI under equal acquisition and candidate budgets?

The contract is deliberately conservative. The implemented substrate covers scene-level oracle RRI, immutable VIN-style offline stores, candidate generation, one-step scoring, early rollout scaffolding, and Rerun inspection. The prerequisite evidence protocol covers M1 data/oracle correctness, V0/V1 target contracts,

invalidity masks/reasons, Zarr-first rollout/Q storage, LRZ-scale generation gates, scene-level splits, and exact coverage reporting. The hard quantitative core is observed target selection, mixed candidate sets, target RRI labels, a target-conditioned one-step scorer, trusted oracle rollouts, and mandatory Q_H . Online discrete Q_H , IQL, actor-critic, continuous control, SceneScript, 3DGS simulators, and real-device guidance are bridge or future-work surfaces unless the M5 evidence justifies escalation.

3 Related Work and Positioning

The local paper manifest and shared bibliography were treated as a design corpus, not as a citation list to pad the proposal. The scientific pattern that emerges is consistent: older active-perception and view-planning work motivates action-conditioned sensing; modern NBV work splits between continuous coverage-driven policies, quality-driven finite candidate ranking, projection shortlists, and radiance-field uncertainty; offline RL contributes support and overestimation warnings for finite-candidate value learning.

Corpus family	Main technical signal	ARIA-NBV use
Active perception and classical view planning [AWB88, Baj88, Ban+00, SRR03]	Viewpoint choice changes what constraints are observable; candidate generation and feasibility are first-class parts of perception.	Keep a finite candidate table, explicit validity, and acquisition budget instead of hiding planning behind a black-box policy.
Receding, projection, and finite-candidate NBV [Bat+22, Bir+16, Fra+25, Jia+25]	Evaluator cost, horizon, branch factor, and reconstruction-quality labels determine whether deeper search is meaningful.	Use mesh-supervised RRI as the utility label; use projection/frontier ideas only as proposal or diagnostic channels.
Egocentric substrate [Eng+23, Met25, Str+24a]	Project Aria supplies calibrated egocentric streams; ASE adds synthetic scale, GT meshes, OBBs, trajectories, and EVL-style 3D state.	Use predicted/observed OBB and EVL support as actor-visible target state; keep GT meshes/boxes for labels and evaluation only.
Continuous and hierarchical learned NBV [Che+24, Lu+26, Sch+17]	Coverage-driven PPO and hierarchical look-at-then-fly policies become powerful when simulator state, reward, and dynamics are mature.	Use as bridge design and baseline context after finite-candidate RRI evidence shows headroom.
Radiance-field / 3DGS active selection [BLL25, Jeo+26, JLD24, Ker+23, LJD25, Mil+20, Pan+22, Str+24b]	Uncertainty, Fisher information, depth, semantic, dynamic, object-specific, and downstream-task utilities can be logged separately.	Adopt utility-channel separation and target focus; do not replace target RRI with 3DGS uncertainty before calibration.
Offline value learning and sequence planning [Che+21, FMP19, Haa+17, Haa+18, HGS15, JLL21, KHW19, KNL21, Kum+20, Mni+13]	Replay, beam decoding, stochastic sampling without replacement, max-over-action overestimation, and offline support mismatch are the relevant hazards.	Train masked fitted Double-Q Q_H first; keep IQL, sequence decoding, soft/energy policies, and actor-critic control as gated ablations.
Structured semantic scene representations [Ave+24, Xie+25]	Autoregressive scene languages and editable layouts support semantic/global memory and human correction.	Use as future global planner context only after observed target contracts and target RRI are stable.

Table 1: Source-backed literature positioning for the ARIA-NBV thesis scope.

The thesis therefore does not claim novelty through a larger continuous action space. Its scientific contribution is the controlled replacement of proxy utility by target-conditioned reconstruction-quality utility under a finite candidate contract. The closest methodological lineage is

e / uncertainty NBV $\rightarrow$ oracle RRI candidate ranking $\rightarrow$ target RRI $\rightarrow$ bounded rollout $\rightarrow$ masked finite-candidate

This lineage makes three negative claims explicit. First, coverage is a useful diagnostic but not the thesis objective. Second, GT meshes and GT target boxes are oracle assets, not actor inputs. Third, offline/continuous RL methods are not safe shortcuts until rollout support, masks, and evaluation are reliable.

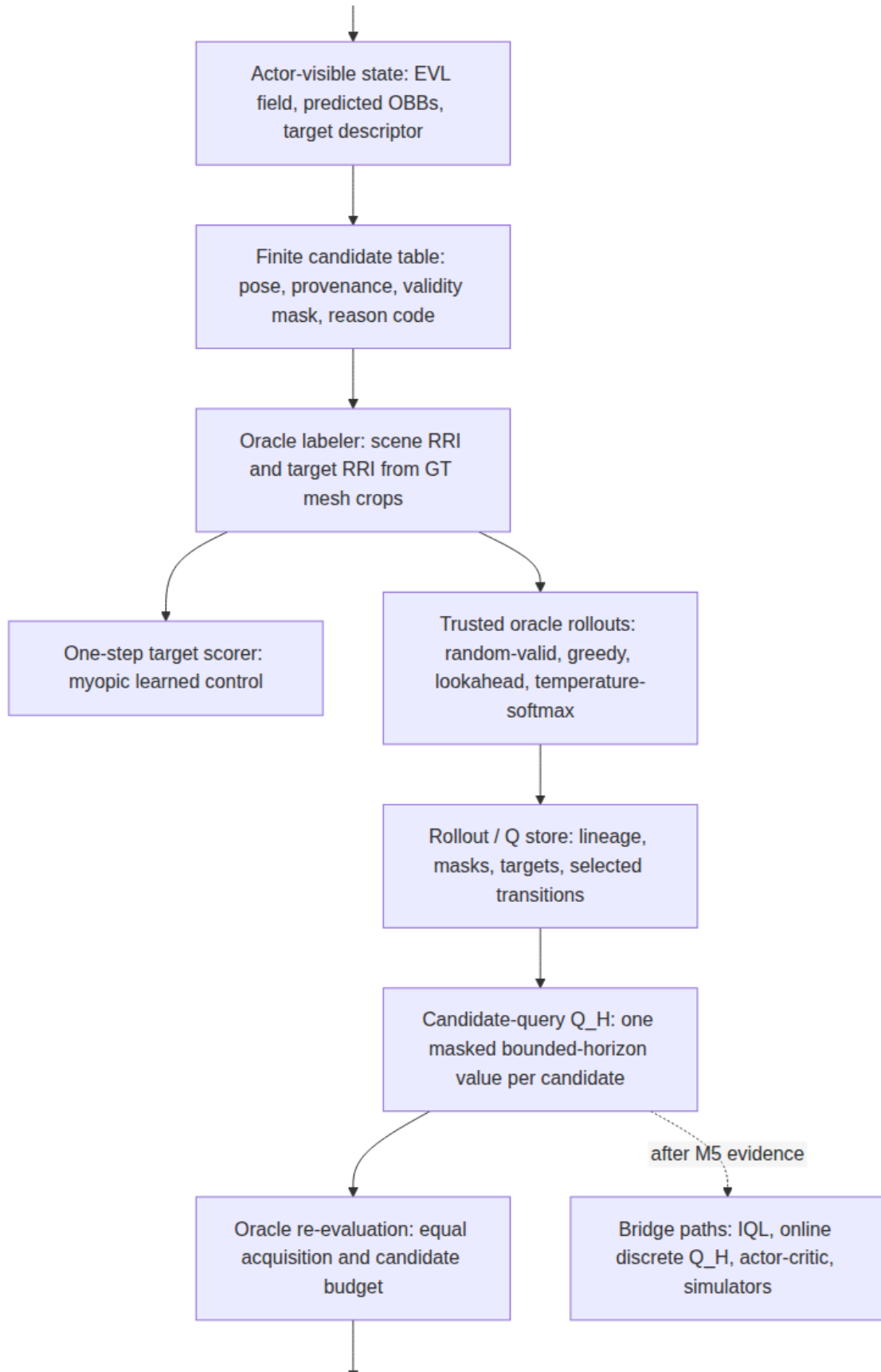


Figure 1: Mermaid flowchart of the thesis evidence chain. Rectangular nodes are thesis-core gates; dashed nodes are bridge paths after Q_H evidence.

4 Objectives and Hypotheses

The proposal has one primary objective: establish a reproducible, target-aware, finite-candidate NBV stack whose selected views improve target reconstruction quality under a fixed acquisition budget. The work is evidence-gated so that a failed substrate does not become an overstated policy claim.

O1: Trustworthy geometry and oracle labels. M1 must demonstrate that offline store indexing, splits, frame conventions, CW90 display handling, candidate-label order, depth rendering/backprojection, invalidity reasons, and Rerun inspection agree on a small trusted subset. The output is not merely a smoke test; it is a public contract that says when RRI labels can be used for training and scale-up.

O2: Target-specific utility without actor leakage. For selected target e , the thesis must compute scene RRI and target RRI separately, using actor-visible observed/predicted target descriptors z_e for selection and model input, but GT crops M_e for labels/evaluation. The first descriptor is predicted/observed OBB center, extents, orientation, class, confidence, projected area, semidense support, EVL support, and relative pose. Crop descriptors and entity tokens are ablations, not initial dependencies.

O3: A learned myopic control. The target-conditioned one-step scorer is the required comparator for planning. It should predict target RRI over the same candidate table and report held-out rank correlation, top- k oracle hit rate, calibration, target visibility, invalid fraction, and failure groups by occlusion, small target, weak descriptor, and candidate infeasibility.

O4: Replayable multi-step data. Rollout data must include random-valid, oracle-greedy/lookahead, and oracle-scored temperature-softmax traces with deterministic replay metadata. Gumbel-Top- k is preferred later diversity evidence, but deterministic greedy/lookahead must be trusted before stochastic branching becomes training data.

O5: Mandatory finite-candidate Q_H . The hard thesis result is a masked candidate-query Transformer trained from ASE oracle rollout traces. It emits one value per candidate:

$$\mathbf{h}_{t,i} = f_{\theta}\left(x_{\text{scene}(s_t^{\text{cf0}})}, x_{\text{target}(z_e)}, x_{\text{hist}(a_{<t}, r_{<t})}, x_{\text{cand}(q_{t,i})}\right), \quad Q_{H,\theta}(s_t^{\text{cf0}}, z_e, q_{t,i}) = W_Q \mathbf{h}_{t,i}.$$

The selected action is

$$a_t^{\theta} = \underset{i:m_{t,i}=1}{\operatorname{argmax}} Q_{H,\theta}(s_t^{\text{cf0}}, z_e, q_{t,i}).$$

Success is measured after oracle re-evaluation of the selected actions, not by the network’s self-reported value. Q_H must beat one-step greedy/model scoring on cumulative target RRI under equal acquisition and candidate budgets, with bounded oracle lookahead reported as the upper bound. If it does not, the thesis reports the failing gate and the smallest defensible target-aware result.

Claim	Metric	Evidence rule
Target utility	J_e^H, G_t^H , scene RRI, log-gain ablation	Endpoint target gain and cumulative return are reported separately; cost curves compare different budgets.
Target input safety	selector rank, OBB match score, target support, leakage checks	V1 actor input uses observed/predicted evidence; GT target crops only produce labels/evaluation.
Candidate space	valid fraction, invalid reasons, target visibility, sampler provenance	All candidates carry hard masks and strategy provenance before scorer or Q_H training.
Planning gain	oracle-evaluated cumulative target RRI under equal budget	Q_H is compared to random-valid, one-step greedy, learned one-step target scorer, and oracle lookahead.
Scale	scenes, snippets, targets, trajectories, seeds, transitions, gaps	Final scale targets all 100 GT-mesh ASE scenes and 4,608 snippet windows, or an exact scene-level held-out subset report.

Table 2: Objective-to-evidence matrix.

5 Method

The method is a sequence of gates. Each gate either produces a trusted artifact for the next gate or records an explicit blocker.

5.1 M1/M2: Data, Geometry, and One-Step Baseline

The first phase freezes the geometry contracts. Candidate poses, rig poses, rendered depths, backprojected point clouds, semidense support, EVL fields, and Rerun diagnostics must agree in frame and indexing. The oracle computes scene-level and target-level labels from

$$\text{CD}(\mathcal{P}, \mathcal{M}_{\text{GT}}) = \mathcal{A}(\mathcal{P}, \mathcal{M}_{\text{GT}}) + \mathcal{C}(\mathcal{P}, \mathcal{M}_{\text{GT}})$$

and

$$\text{RRI}_{e(q)} = \frac{\text{CD}(\mathcal{P}_t^e, M_e) - \text{CD}(\mathcal{P}_t^e \cup \mathcal{P}_q^e, M_e)}{\text{CD}(\mathcal{P}_t^e, M_e) + \varepsilon}$$

without aggressive mesh or point-cloud downsampling in fine-detail runs. The one-step VIN-style scorer then serves as the learned myopic baseline, not as the final policy. Its main purpose is to test whether actor-visible state and candidate features predict oracle RRI well enough to justify planning.

5.2 M3/M4: Target Contract and Target-Conditioned Scoring

Target selection produces a small top- K set of actor-visible observed/predicted targets per snippet. For V1, the selector scores predicted OBB confidence, projected visibility, semidense/EVL support, support deficit, distance, and class eligibility; it refuses GT OBB input. The labeler then matches selected targets to GT crops using class compatibility, OBB overlap, visibility/support, projected area, and semidense/EVL support. Ambiguous matches and the exact near-solved-target threshold remain advisor-facing details.

The target-conditioned one-step scorer receives the same finite candidate set as Q_H and predicts target RRI. Its ablations are bounded: target descriptor choice, surface reconstruction input, CORAL variant, auxiliary regression, candidate-relative pose encoding, and calibration/stage features. This keeps model work from becoming unbounded architecture search before the planning question is answered.

5.3 M5: Bounded Rollout and Q_H Training

For each root state, deterministic oracle one-step greedy and bounded oracle lookahead are generated first. Let B be branch factor, H horizon, and $\mathcal{A}(s_t) = \{i : m_{t,i} = 1\}$. Greedy selection is

$$a_t^{\text{greedy}} = \underset{i \in \mathcal{A}(s_t)}{\operatorname{argmax}} r_t^{e(i)}.$$

Bounded lookahead selects the first action of the best rollout:

$$a_t^{\text{look}} = \pi_0 \left(\underset{a_{t:t+H-1}}{\operatorname{argmax}} \sum_{k=0}^{H-1} \gamma^k r_{t+k}^e \right),$$

where $\pi_0(\cdot)$ extracts the first action. Temperature-softmax traces then add support:

$$P(a_t = i \mid s_t) = \frac{\exp(\beta u_{t,i})}{\sum_{j:m_{t,j}=1} \exp(\beta u_{t,j})},$$

where $u_{t,i}$ may be oracle immediate RRI, oracle lookahead value, or a trusted scorer. The rollout store preserves logits, probabilities, entropy, temperature, selected action, RNG seed, target identity, candidate provenance, and invalid reasons.

Q_H is trained as masked fitted Double-Q over finite candidates. With online parameters θ and target parameters $|\langle \theta \rangle$,

$$i^* = \operatorname{argmax}_{i:m_{t+1,i}=1} Q_{H,\theta}(s_{t+1}^{\text{cf0}}, z_e, q_{t+1,i}),$$

$$y_t = r_t^e + \gamma Q_{H,|\langle \theta \rangle}(s_{t+1}^{\text{cf0}}, z_e, q_{t+1,i^*}),$$

$$\mathcal{L}_{Q(\theta)} = \frac{1}{|\mathcal{D}|} \sum_{(s,a,r,s') \in \mathcal{D}} m_{t,a} (Q_{H,\theta}(s_t^{\text{cf0}}, z_e, q_{t,a}) - y_t)^2.$$

This target family uses the Double-DQN selector/evaluator separation to reduce max-operator overestimation [HGS15]. CQL, BCQ, IQL, Decision Transformer, soft/energy policies, PPO, and SAC are comparison references or later ablations because they require stronger support assumptions, sequence decoding, or an online simulator [Che+21, FMP19, Haa+17, Haa+18, KNL21, Kum+20, Sch+17].

5.4 Evaluation

All selected actions are oracle-evaluated under the same budget. The main table compares random-valid, one-step oracle greedy, learned one-step target scorer, bounded oracle lookahead, temperature-softmax rollouts, and learned Q_H on scene-level splits. Required columns are endpoint target gain J_e^H , cumulative target return G_0^H , scene RRI, view count, path length, invalid-action rate, runtime, scenes, snippets, targets, trajectories, rollout seeds, transitions, and coverage gaps. Quality-cost curves are used when budgets differ.

Gate	Artifact	Fail-safe
M1	contract report, Rerun examples, throughput and frame checks	block target/RL scale-up if geometry or labels are not trustworthy
M3	V0/V1 target RRI and observed-only target selector	fall back to target oracle diagnostics if actor-visible matching is sparse
M4	target-conditioned one-step scorer	use as myopic baseline even if accuracy is modest, provided calibration is reported
M5	rollout store, oracle lookahead, and candidate-query Q_H	report the exact blocker if Q_H does not beat myopic controls
M6+	online/IQL/actor-critic/simulator bridge design	do not promote bridge work to thesis core without M5 evidence

Table 3: Method gates and fail-safe interpretation.

6 Schedule and Risk Control

The planned thesis window runs from 29 April 2026 to 30 September 2026. The schedule is organized around evidence gates rather than feature optimism: offline-store trust and oracle correctness precede target-conditioned modeling; target RRI and observed target selection precede Q_H data generation; bridge work begins only after the finite-candidate planning evidence is interpretable.

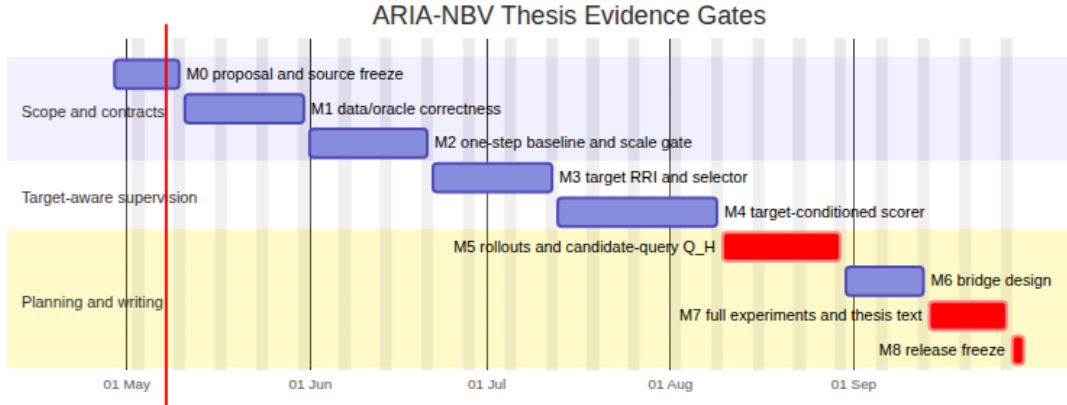


Figure 2: Mermaid Gantt chart for the proposal timeline and evidence gates.

Dates	Milestone	Exit condition
2026-04-29 to 2026-05-10	M0 scope/proposal	Proposal, roadmap, questions, source policy, and hard Q_H boundary are aligned and renderable.
2026-05-11 to 2026-05-31	M1 data/oracle	Offline-store, split, frame/CW90, candidate-label, depth/backprojection, Rerun, and throughput evidence pass or block scale-up.
2026-06-01 to 2026-06-21	M2 one-step baseline	Scene-level VIN baseline, calibration/ranking plots, LRZ sharding plan, and Zarr rollout/Q schema are ready.
2026-06-22 to 2026-07-12	M3 target RRI	V0 GT-OBB target labels and V1 OBS-SEL / PRED-Q / GT-EVAL contract are trusted on a small subset.
2026-07-13 to 2026-08-09	M4 target scorer	Target-conditioned one-step scorer is evaluated against scene-level scoring and oracle target RRI.
2026-08-10 to 2026-08-30	M5 rollouts/ Q_H	Random-valid, oracle-greedy/lookahead, and temperature-softmax traces are stable; candidate-query Q_H is trained and oracle-evaluated.
2026-08-31 to 2026-09-13	M6 bridge design	Online discrete Q_H , IQL, actor-critic, hierarchy, and simulator bridge are written as designs or gated ablations.
2026-09-14 to 2026-09-27	M7 experiments/writing	Final tables, figures, failure cases, coverage report, and thesis narrative are frozen.
2026-09-28 to 2026-09-30	M8 release	Configs, smoke checks, demo path, and final PDF artifacts are reproducible.

Table 4: Milestone exit criteria.

The most likely failure mode is not that continuous RL is needed too early; it is that target labels, validity masks, or scale generation are not trusted enough for learned planning. The fallback policy is therefore conservative. If M1 fails, the thesis remains a geometry/oracle contract and one-step scorer study. If M3 target matching is sparse, target RRI remains an oracle diagnostic while scene-level scoring is reported honestly. If Q_H fails to beat myopic controls, the thesis reports whether the limiting factor is candidate support, target observability, reward definition, or model capacity. None of these fallbacks reclassifies the mandatory Q_H attempt as optional.

7 Preliminary Thesis Outline

The thesis will be written as an empirical methods thesis. The chapter order mirrors the validation order so that claims are introduced only after the corresponding contract is established.

Chapter	Purpose	Expected evidence
Introduction	Motivate target-conditioned, quality-driven NBV for egocentric indoor reconstruction and state the finite-candidate thesis claim.	Research questions, contribution boundary, and source-backed scope.
Background	Review active perception, NBV, RRI, ASE/Project Aria, EFM3D/EVL, target-aware 3DGS, and offline value learning.	Literature synthesis with adoption/rejection decisions.
Data and geometry contracts	Describe snippets, calibration, frames, offline stores, candidates, rendered depths, backprojection, masks, and Rerun inspection.	M1 contract report, visual diagnostics, throughput, and known limitations.
Oracle RRI and target RRI	Define scene/target distances, crop matching, invalidity, and label generation.	Label distributions, target crops, target/scene divergence, and failure cases.
Target-conditioned scoring	Present actor-visible target encoding, candidate features, VIN-style model, ordinal/regression losses, and calibration.	Held-out ranking, top- k oracle hit, ablations, calibration, and target-specific failures.
Bounded rollout and Q_H	Compare random-valid, one-step greedy, learned one-step scorer, oracle lookahead, temperature-softmax traces, and candidate-query Q_H .	Cumulative target RRI, endpoint target gain, scene RRI, cost, invalidity, runtime, and rollout visualizations.
Discussion and conclusion	Interpret limits, scale blockers, simulator gaps, semantic/global planning, and real-device bridge paths.	Evidence-gated conclusion and reproducibility package.

Table 5: Preliminary thesis chapter outline.

The expected final contribution is a reproducible target-aware finite-candidate view-selection study, not a broad claim that continuous reinforcement learning has been solved for egocentric reconstruction.

Bibliography

- [Baj88] R. Bajcsy, “Active Perception,” *Proceedings of the IEEE*, vol. 76, no. 8, pp. 966–1005, 1988, doi: 10.1109/5.5968.
- [AWB88] J. Aloimonos, I. Weiss, and A. Bandyopadhyay, “Active Vision,” *International Journal of Computer Vision*, vol. 1, no. 4, pp. 333–356, 1988, doi: 10.1007/BF00133571.
- [SRR03] W. R. Scott, G. Roth, and J.-F. Rivest, “View Planning for Automated Three-Dimensional Object Reconstruction and Inspection,” *ACM Computing Surveys*, vol. 35, no. 1, pp. 64–96, 2003, doi: 10.1145/641865.641868.
- [Fra+25] N. Frahm *et al.*, “VIN-NBV: A View Introspection Network for Next-Best-View Selection.” [Online]. Available: <https://arxiv.org/abs/2505.06219>
- [Eng+23] J. Engel *et al.*, “Project Aria: A New Tool for Egocentric Multi-Modal AI Research.” [Online]. Available: <https://arxiv.org/abs/2308.13561>
- [Met25] Meta Platforms Inc., “Aria Synthetic Environments Dataset.” [Online]. Available: https://facebookresearch.github.io/projectaria_tools/docs/open_datasets/aria_synthetic_environments_dataset
- [Str+24] J. Straub, D. DeTone, T. Shen, N. Yang, C. Sweeney, and R. Newcombe, “EFM3D: A Benchmark for Measuring Progress Towards 3D Egocentric Foundation Models.” [Online]. Available: <https://arxiv.org/abs/2406.10224>
- [Che+24] X. Chen, Q. Li, T. Wang, T. Xue, and J. Pang, “GenNBV: Generalizable Next-Best-View Policy for Active 3D Reconstruction,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16436–16445. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Chen_GenNBV_Generalizable_Next-Best-View_Policy_for_Active_3D_Reconstruction_CVPR_2024_paper.html

- [Lu+26] C.-Y. Lu *et al.*, “Hestia: Voxel-Face-Aware Hierarchical Next-Best-View Acquisition for Efficient 3D Reconstruction.” [Online]. Available: https://johnnylu305.github.io/hestia_web
- [Pan+22] X. Pan, Z. Lai, S. Song, and G. Huang, “ActiveNeRF: Learning Where to See with Uncertainty Estimation.” [Online]. Available: <https://arxiv.org/abs/2209.08546>
- [JLD24] W. Jiang, B. Lei, and K. Daniilidis, “FisherRF: Active View Selection and Uncertainty Quantification for Radiance Fields using Fisher Information.” [Online]. Available: <https://arxiv.org/abs/2311.17874>
- [Str+24] M. Strong, B. Lei, A. Swann, W. Jiang, K. Daniilidis, and M. K. III, “Next Best Sense: Guiding Vision and Touch with FisherRF for 3D Gaussian Splatting.” [Online]. Available: <https://arxiv.org/abs/2410.04680>
- [LJD25] Y. Li, W. Jiang, and K. Daniilidis, “Next Best View Selections for Semantic and Dynamic 3D Gaussian Splatting.” [Online]. Available: <https://arxiv.org/abs/2512.22771>
- [Jeo+26] S. Jeong, E. Lee, J. Kim, and A. Kim, “Informative Object-centric Next Best View for Object-aware 3D Gaussian Splatting in Cluttered Scenes.” [Online]. Available: <https://arxiv.org/abs/2602.08266>
- [BLL25] W. G. Bae, S. Lee, and J. T. Lee, “Finding Optimal Viewpoints for Monocular 3D Human Pose Estimation in Dynamic 3D Gaussian Splatting Space,” in *Proceedings of the IEEE International Conference on Advanced Video and Signal-Based Surveillance*, 2025. doi: 10.1109/AVSS65446.2025.11149906.
- [Ban+00] J. E. Banta, L. R. Wong, C. Dumont, and M. A. Abidi, “A Next-Best-View System for Autonomous 3-D Object Reconstruction,” *IEEE Transactions on Systems, Man, and Cybernetics - Part A: Systems and Humans*, vol. 30, no. 5, pp. 589–598, 2000, doi: 10.1109/3468.867866.
- [Bir+16] A. Bircher, M. Kamel, K. Alexis, H. Oleynikova, and R. Siegwart, “Receding Horizon “Next-Best-View” Planner for 3D Exploration,”

- in *2016 IEEE International Conference on Robotics and Automation*, 2016, pp. 1462–1468. doi: 10.1109/ICRA.2016.7487281.
- [Bat+22] A. Batinovic, A. Ivanovic, T. Petrovic, and S. Bogdan, “A Shadowcasting-Based Next-Best-View Planner for Autonomous 3D Exploration,” *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 2969–2976, 2022, doi: 10.1109/LRA.2022.3146586.
- [Jia+25] Z. Jia, Y. Li, Q. Hao, and S. Zhang, “PB-NBV: Efficient Projection-Based Next-Best-View Planning Framework for Reconstruction of Unknown Objects,” *IEEE Robotics and Automation Letters*, vol. 10, no. 7, pp. 7444–7451, 2025, doi: 10.1109/LRA.2025.3573631.
- [Sch+17] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal Policy Optimization Algorithms.” [Online]. Available: <https://arxiv.org/abs/1707.06347>
- [Mil+20] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, “NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis,” in *Computer Vision - ECCV 2020*, 2020, pp. 405–421. doi: 10.1007/978-3-030-58452-8_24.
- [Ker+23] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis, “3D Gaussian Splatting for Real-Time Radiance Field Rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023, doi: 10.1145/3592433.
- [JLL21] M. Janner, Q. Li, and S. Levine, “Offline Reinforcement Learning as One Big Sequence Modeling Problem.” [Online]. Available: <https://arxiv.org/abs/2106.02039>
- [KHW19] W. Kool, H. van Hoof, and M. Welling, “Stochastic Beams and Where to Find Them: The Gumbel-Top-k Trick for Sampling Sequences Without Replacement,” in *Proceedings of the 36th International Conference on Machine Learning*, 2019. [Online]. Available: <https://arxiv.org/abs/1903.06059>
- [Mni+13] V. Mnih *et al.*, “Playing Atari with Deep Reinforcement Learning,” *CoRR*, 2013, [Online]. Available: <http://arxiv.org/abs/1312.5602>

- [HGS15] H. van Hasselt, A. Guez, and D. Silver, “Deep Reinforcement Learning with Double Q-learning.” [Online]. Available: <https://arxiv.org/abs/1509.06461>
- [Kum+20] A. Kumar, A. Zhou, G. Tucker, and S. Levine, “Conservative Q-Learning for Offline Reinforcement Learning,” in *Advances in Neural Information Processing Systems*, 2020, pp. 1179–1191. [Online]. Available: <https://papers.nips.cc/paper/2020/hash/0d2b2061826a5df3221116a5085a6052-Abstract.html>
- [FMP19] S. Fujimoto, D. Meger, and D. Precup, “Off-Policy Deep Reinforcement Learning without Exploration,” in *Proceedings of the 36th International Conference on Machine Learning*, in Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 2052–2062. [Online]. Available: <https://proceedings.mlr.press/v97/fujimoto19a.html>
- [Che+21] L. Chen *et al.*, “Decision Transformer: Reinforcement Learning via Sequence Modeling,” in *Advances in Neural Information Processing Systems*, 2021, pp. 15084–15097. [Online]. Available: https://papers.nips.cc/paper_files/paper/2021/hash/7f489f642a0ddb10272b5c31057f0663-Abstract.html
- [KNL21] I. Kostrikov, A. Nair, and S. Levine, “Offline Reinforcement Learning with Implicit Q-Learning.” [Online]. Available: <https://arxiv.org/abs/2110.06169>
- [Haa+17] T. Haarnoja, H. Tang, P. Abbeel, and S. Levine, “Reinforcement Learning with Deep Energy-Based Policies,” in *Proceedings of the 34th International Conference on Machine Learning*, 2017. [Online]. Available: <https://arxiv.org/abs/1702.08165>
- [Haa+18] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor,” in *Proceedings of the 35th International Conference on Machine Learning*, 2018, pp. 1861–1870. [Online]. Available: <https://arxiv.org/abs/1801.01290>

- [Ave+24] A. Avetisyan *et al.*, “SceneScript: Reconstructing Scenes With An Autoregressive Structured Language Model.” [Online]. Available: <https://arxiv.org/abs/2403.13064>
- [Xie+25] C. Xie *et al.*, “Human-in-the-Loop Local Corrections of 3D Scene Layouts via Infilling.” [Online]. Available: <https://arxiv.org/abs/2503.11806>